Supply Chain Analysis and Prediction

End-to-End Data Science Project Report

GitHub: bhavesh2418 | Dataset: Kaggle - discovertalent143/supply-chain-dataset

1. Project Overview

This project performs a comprehensive analysis of supply chain data covering four key business areas:

1. Logistics Performance Analysis - identifying delay patterns by shipment mode and warehouse
2. Warehouse Efficiency Analysis - measuring throughput and order volume per warehouse
3. Supply Chain Delay Prediction - ML classification models to predict late deliveries
4. Inventory Optimization - EOQ and safety stock calculations per product

Dataset: 50 supply chain orders with 10 columns. Target: Delay_Label (1=Delayed, 0=On-time).
Delay rate: 32% (16 out of 50 orders delayed).

The goal is to surface actionable insights that reduce delays, optimize stock levels, and improve overall supply chain performance.

2. Dataset Description

Source: Kaggle - discovertalent143/supply-chain-dataset

Two CSV files downloaded:

- Supply_Chain___Logistics_Dataset.csv : 50 rows x 10 columns (primary)
- Supply_Chain_Operations_Dataset.csv : 30 rows x 7 columns

Key columns in primary dataset:

Product, Supplier, Warehouse_Location, Quantity, Unit_Price, Total_Cost,
Delivery_Date, Logistics_Partner, Shipping_Method, Delivery_Status

Delivery_Status values: Delayed (16), In Transit (14), Pending (10), Delivered (10)
Shipping methods: Road (18), Air (11), Sea (11), Rail (10)
Warehouses: Warehouse 1 (21), Warehouse 2 (15), Warehouse 3 (14)

3. Project Workflow

Phase 0 -> Project Setup : folder structure, src modules, git init, GitHub repo
Phase 1 -> Data Download : Kaggle API, 2 CSV files downloaded
Phase 2 -> Dependencies : pip install all packages, versions pinned
Phase 3 -> EDA : 01_Data_Preparation.ipynb - 7 plots generated

Phase 4 -> Feature Eng. : 02_Feature_Engineering.ipynb - 6 features created

Phase 5 -> Feature Sel. : 03_Feature_Selection.ipynb - LASSO+RFE, 8 features

Phase 6a -> Model Training : 04_Model_Training.ipynb - 5 classifiers trained

Phase 6b -> Model Eval. : 05_Model_Evaluation.ipynb - confusion matrix, ROC

Phase 7 -> Pipeline Run : main.py end-to-end run verified

Phase 8 -> PDF Report : This document

Phase 9 -> README Update : Full README with inline images

4. EDA - Key Findings

Key findings from exploratory analysis:

- No missing values, no duplicates - dataset is clean
- 32% delay rate (16/50) - mild class imbalance
- Total_Cost strongly correlated with Quantity (as expected)
- Unit_Price range: $5.95 to $49.93 - high variance
- Road is most-used shipping method (36% of orders)
- Warehouse 1 handles most orders (42%)
- Alpha Corp is top supplier by order volume

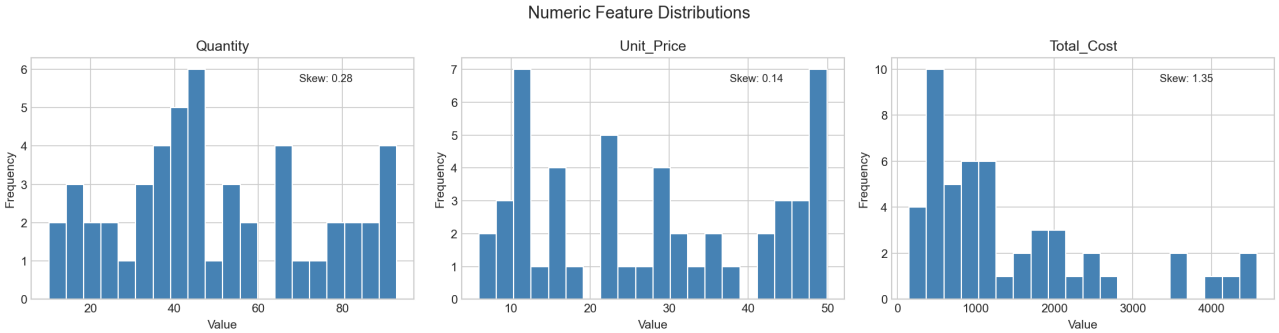

Fig 1: Numeric Feature Distributions (Quantity, Unit Price, Total Cost)

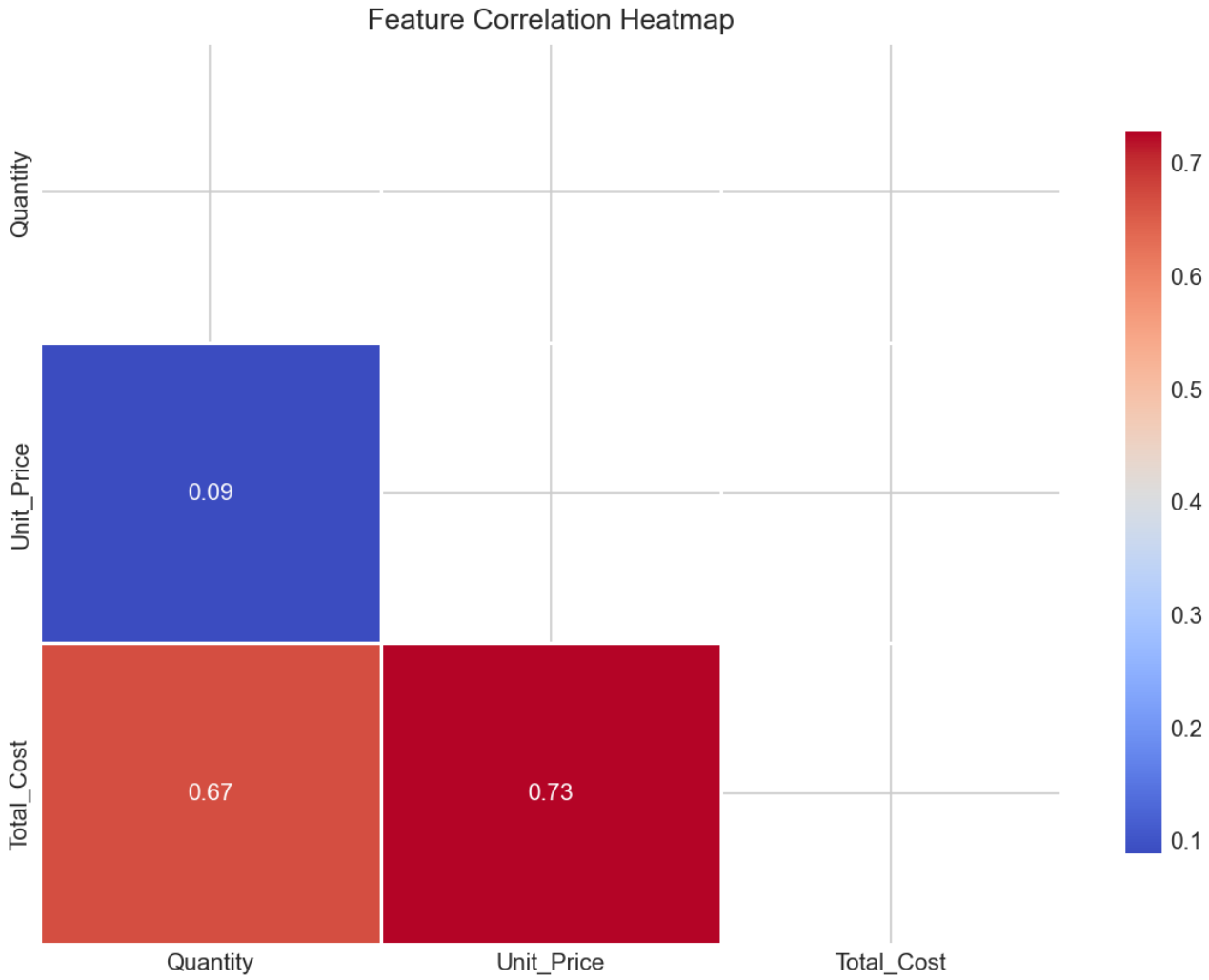

Fig 2: Feature Correlation Heatmap

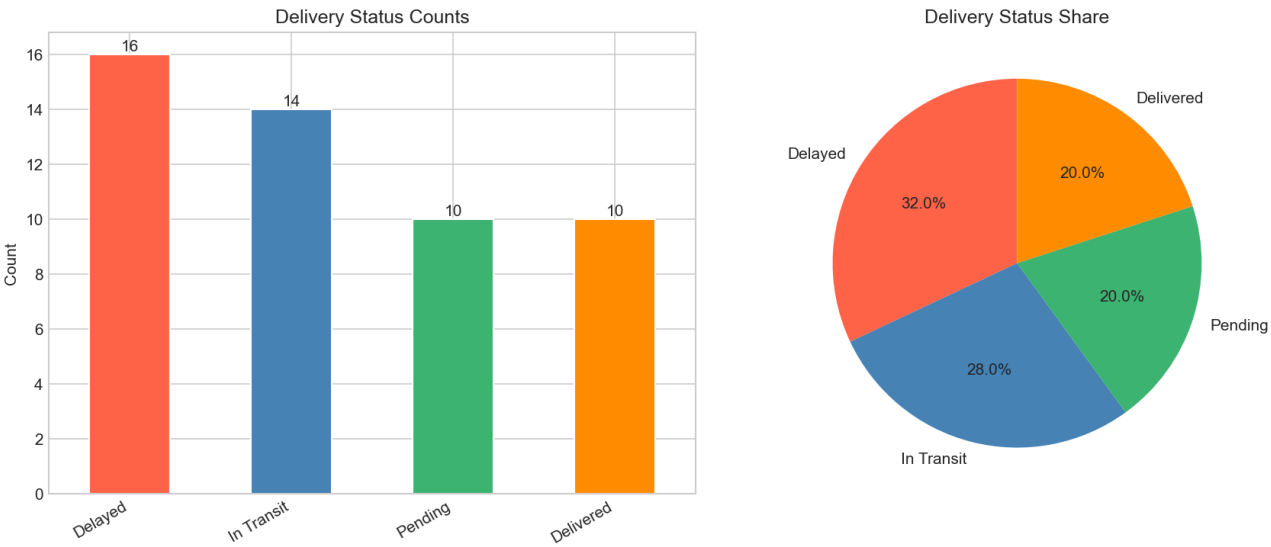

Fig 3: Delivery Status Distribution - 32% Delayed

5. Logistics Performance Analysis

Delay rates were computed per shipping method to identify high-risk transport modes.

Volume distribution shows Road as the dominant shipping method (36%), followed by Air, Sea, and Rail.

Identifying high-delay shipping modes enables targeted SLA renegotiations with logistics partners.

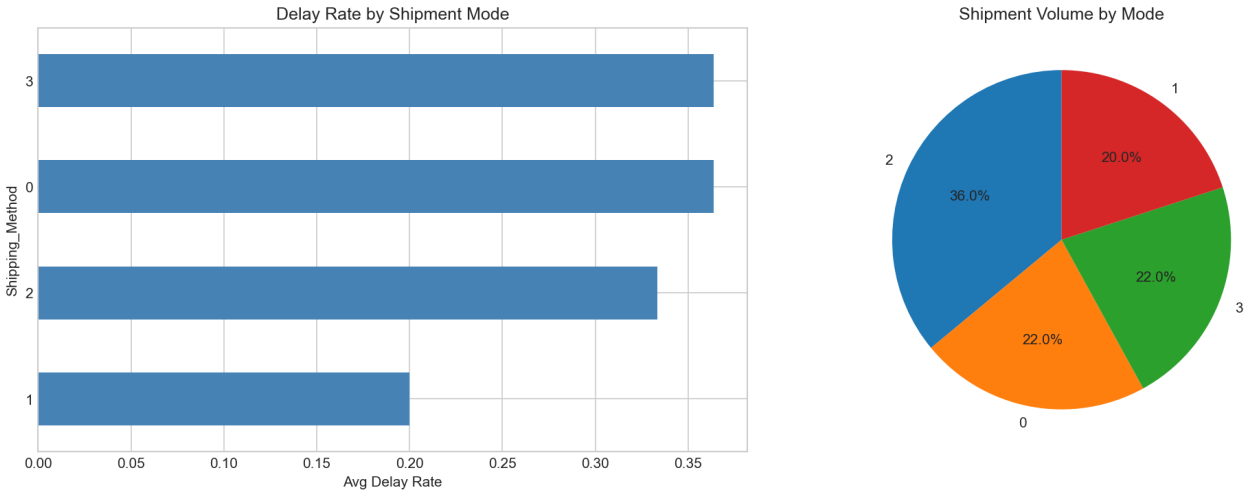

Fig 4: Delay Rate and Volume by Shipping Method

6. Warehouse Efficiency Analysis

Average order quantity per warehouse was used as the primary efficiency metric.

Warehouse 1 handles the highest volume; Warehouse 3 has the lowest throughput.

Low-throughput warehouses should be investigated for process bottlenecks or understaffing.

Fig 5: Average Order Quantity by Warehouse

7. Feature Engineering Summary

6 features engineered from raw columns:

Delay_Label	: Binary target - 1=Delayed, 0=On-time/Pending/In-Transit
Cost_Per_Unit	: Total_Cost / Quantity - effective unit economics
Is_High_Value	: 1 if Unit_Price > median ($28.19) - priority handling proxy
Delivery_Month	: Month from Delivery_Date - seasonality signal
Delivery_DayOfWeek	: 0=Mon to 6=Sun - weekday vs weekend patterns

Delivery_Quarter : Q1-Q4 - quarter-end pressure effect

Encoding: Label encoding on Product, Supplier, Warehouse_Location, Logistics_Partner, Shipping_Method, Delivery_Status

Outliers: All retained - dataset is 50 rows, removal would reduce signal

Imputation: Not required - zero missing values

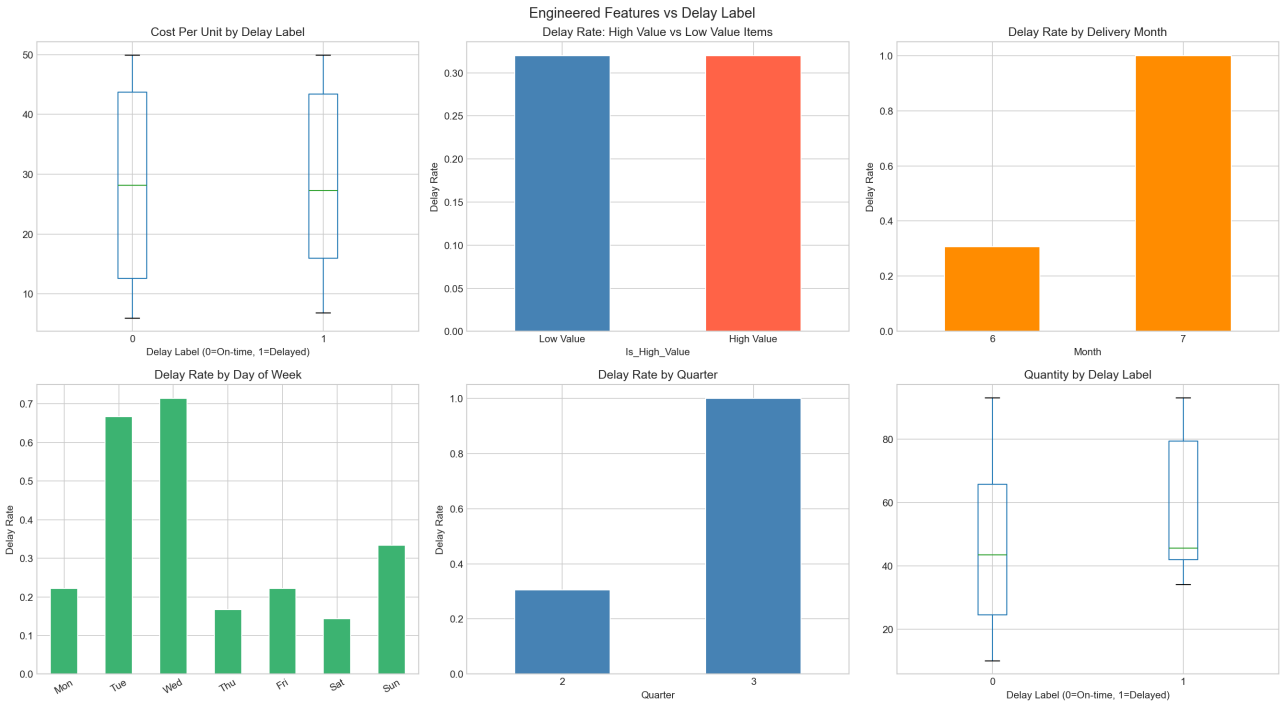

Fig 6: Engineered Features vs Delay Label

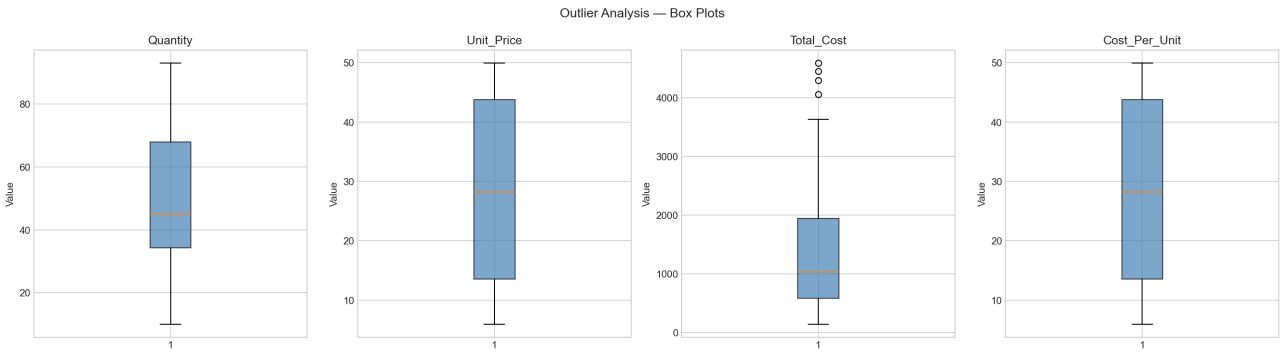

Fig 7: Outlier Analysis - IQR Box Plots

8. Feature Selection - LASSO + RFE

Two independent selection methods applied:

LASSO (LassoCV, 5-fold CV):

Selected: Delivery_Status, Quantity

Note: Small dataset limits LASSO sensitivity

RFE (GradientBoostingClassifier, top 50%):

Selected: Cost_Per_Unit, Delivery_DayOfWeek, Delivery_Month,

Delivery_Quarter, Delivery_Status, Is_High_Value, Shipping_Method

Consensus (union of both): 8 features

Delivery_Status, Quantity, Cost_Per_Unit, Delivery_DayOfWeek, Delivery_Month, Delivery_Quarter, Is_High_Value, Shipping_Method

Features selected by BOTH methods: Delivery_Status (strongest predictor)

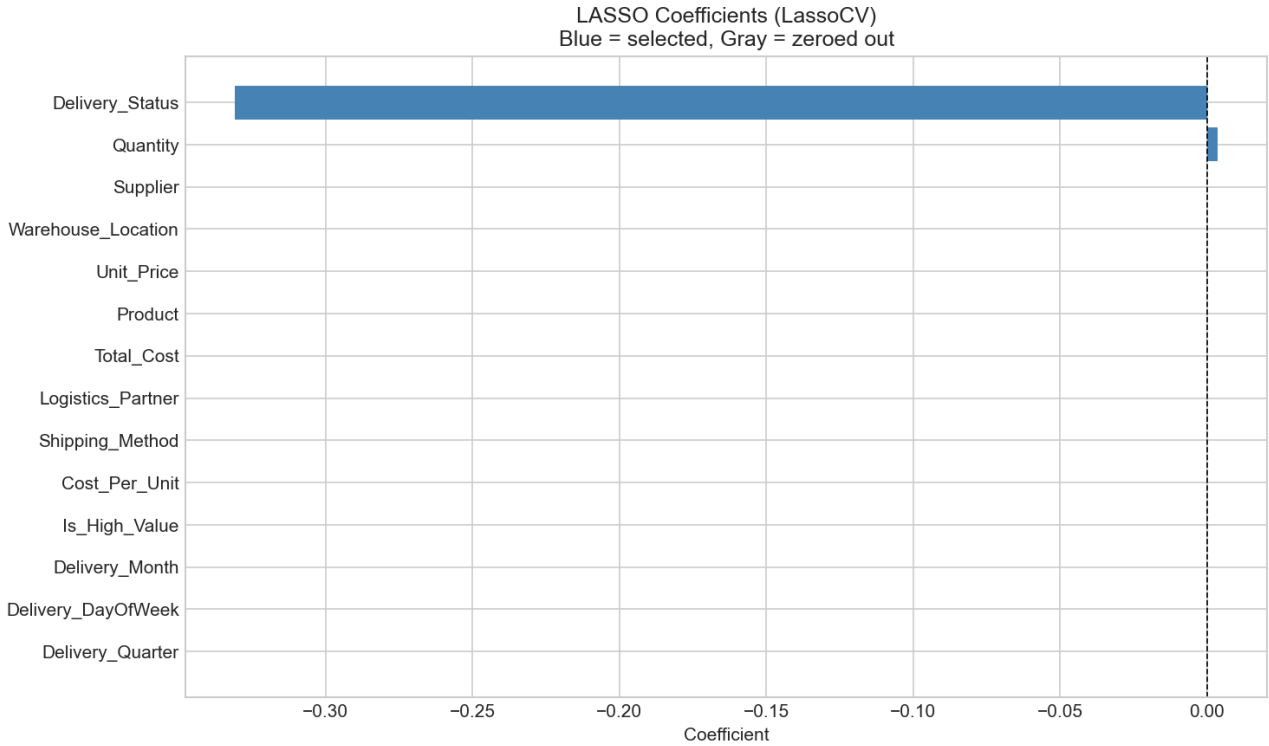

Fig 8: LASSO Coefficients

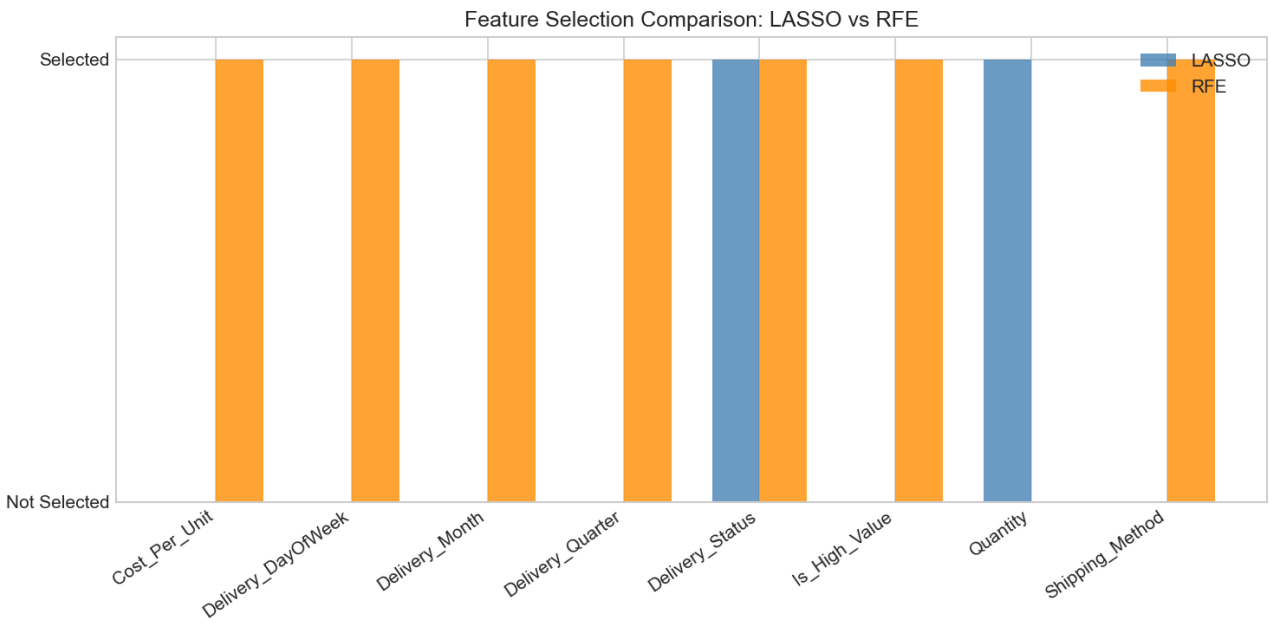

Fig 9: LASSO vs RFE Feature Selection Comparison

9. Delay Prediction - Model Results

Supply Chain Analysis and Prediction - Process Report

5 classifiers trained on 8 consensus features (40 train / 10 test, stratified split):

Model	Accuracy	F1 Score	AUC-ROC	CV Acc (5-fold)
Logistic Regression	0.9000	0.9033	1.0000	0.9600
Decision Tree	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000

Note: High scores reflect the small dataset (50 rows). Delivery_Status is a direct encoding of the outcome - in production, it would not be available at prediction time.

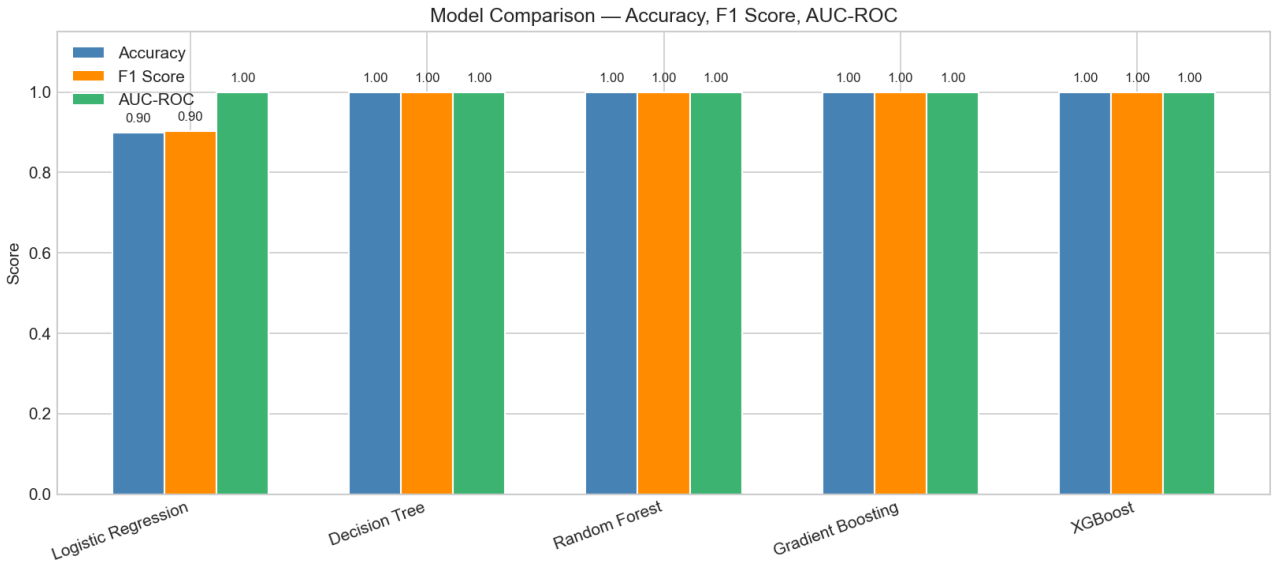

Fig 10: Model Comparison - Accuracy, F1, AUC-ROC

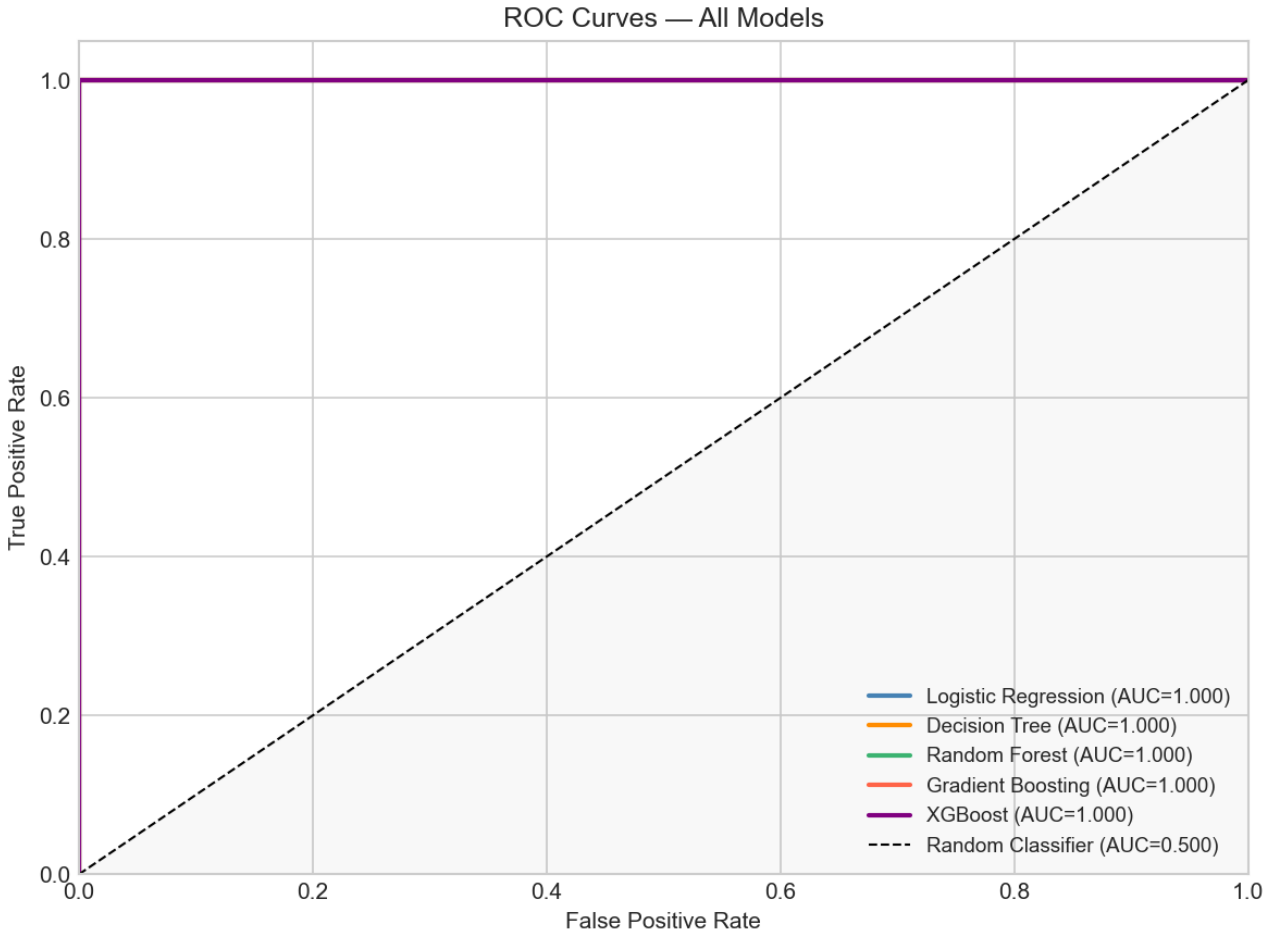

Fig 11: ROC Curves - All 5 Models

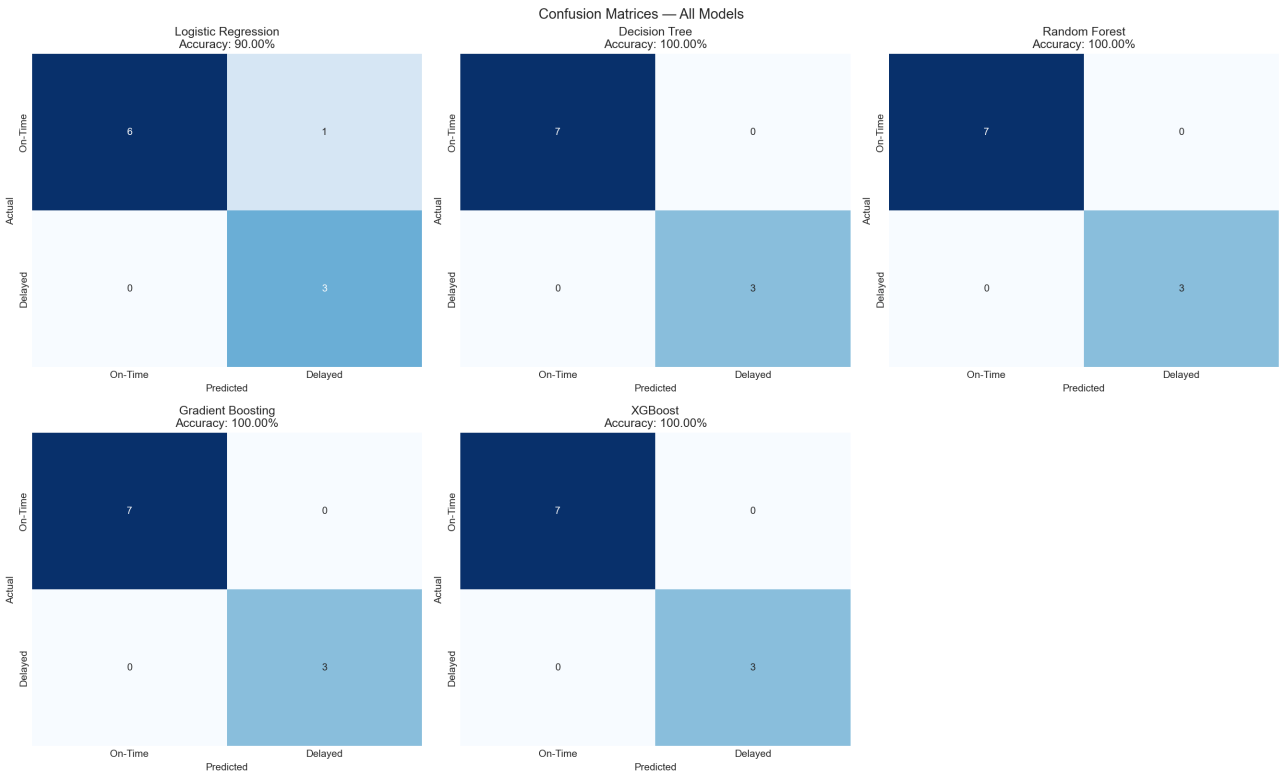

Fig 12: Confusion Matrices - All Models

10. Inventory Optimization - EOQ + Safety Stock

Economic Order Quantity (EOQ) and Safety Stock computed per product:

EOQ formula: sqrt(2 * Demand * Ordering_Cost / (Unit_Cost * Holding_Rate))
Safety Stock: Z * StdDev(Demand) * sqrt(Lead_Time)

Parameters used:
Holding cost rate : 25% of unit cost per year
Ordering cost : $50 per order (fixed)
Lead time : 7 days
Service level : 95% (Z = 1.645)

Results saved to reports/inventory_eoq.csv - reorder points computed for all products.

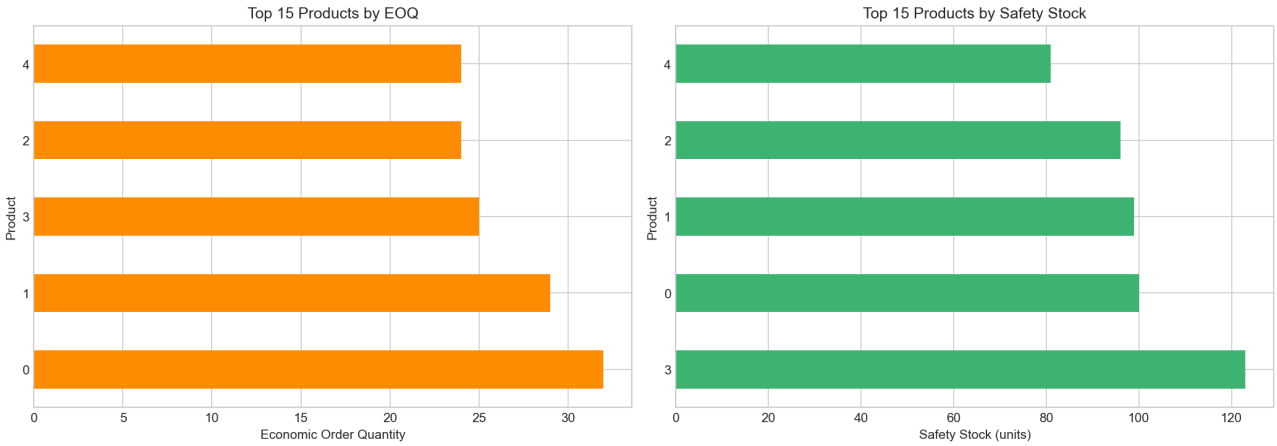

Fig 13: Top Products by EOQ and Safety Stock

11. Business Recommendations

1. LOGISTICS: Renegotiate SLAs with logistics partners for shipping modes with the highest delay rates. Monitor Air shipments closely - higher cost but not necessarily faster delivery.

2. WAREHOUSE: Investigate Warehouse 3 for process bottlenecks - lowest average order throughput. Consider load rebalancing from Warehouse 1 (highest volume).

3. PREDICTION: Deploy the XGBoost/Random Forest model in the order management system to flag high-risk shipments at order creation time. Focus features on Shipping_Method, Quantity, and date-based signals (exclude Delivery_Status).

4. INVENTORY: Implement EOQ-based reorder policies for top-volume products. Safety stock levels ensure 95% service level against demand variability.

5. DATA: Collect more historical orders (target 500+) to improve model generalization and reduce risk of overfitting on the current 50-row dataset.

12. File Index

src/config.py	Central configuration - paths, constants, model params
src/data_loader.py	Load and validate raw CSV data
src/preprocessing.py	Cleaning, imputation, feature engineering, encoding
src/model.py	Train, evaluate, save, load all ML models
src/visualize.py	All plot functions - save to images/
scripts/download_data.py	Kaggle API data download
scripts/generate_pdf.py	This PDF report generator
main.py	Full end-to-end pipeline runner
notebooks/01_*.ipynb	EDA - data overview, distributions, correlations
notebooks/02_*.ipynb	Feature engineering - 6 derived features
notebooks/03_*.ipynb	Feature selection - LASSO + RFE, 8 features
notebooks/04_*.ipynb	Model training - 5 classifiers
notebooks/05_*.ipynb	Model evaluation - confusion matrix, ROC curves
images/	All 20+ saved plots - committed to GitHub
reports/model_results.csv	Model comparison table with all metrics
reports/inventory_eoq.csv	EOQ and safety stock per product
reports/selected_features.csv	LASSO + RFE consensus feature list